MCS-324	UNIX NETWORK PROGRAMMING	L T P 3 1 0	4 Credits

Course Objectives:

CO1	Able to learn about the protocols which are using in the current scenario.
CO2	To learn and understand client server relations and OSI programming Implementation of the socket and IPC
CO3	Develop skills in network programming techniques.
CO4	Implement network services that communicate through the Internet.
CO5	Apply the client-server model in networking applications.

MODULE I

Client/Server Model, Peer-to-Peer Model, overview of IPv4 and IPv6, TCP and UDP,

MODULE II

Socket programming, Multiplexing I/O, Encapsulation, Unix Domain Protocols, Daemon Processes, super server, broadcasting and Multicasting,

MODULE III

Threaded network programming, Raw Socket, HTTP Server Design.

References:

1. *W. Richard Stevens, "UNIX Network Programming," Volume 1, second edition, Prentice Hall. ISBN #0-13-490012-X.*
2. *Douglas Comer, "Internetworking with TCP/IP," Volume I, II & III, Prentice Hall.*

Course Outcome:

CO1	Explain OSI Model and Standard Internet Services and Protocols
CO2	How to handle server process termination
CO3	Acquire the knowledge of Elementary TCP sockets and I/O Multiplexing and socket options
CO4	Demonstrate the concepts of fifos streams messages and Remote logins
CO5	Apply knowledge of Unix operating systems to build robust client and server software for this environment;